

Enhancing Freehand VR Interaction Using Fingertip Deformation on User Performance

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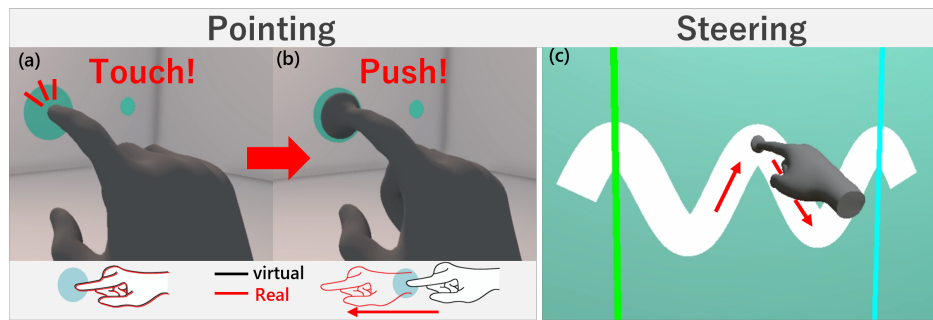


Figure 1: Fingertip deformation feedback in VR. (a) The virtual hand touches a blue UI button without deformation. (b) Button press triggers fingertip deformation, providing depth information and pseudo-haptics feedback. (c) Steering task where users track a wavy path using fingertip deformations to maintain appropriate depth levels.

Abstract

This study investigated the use of the deformation of the fingertips of a virtual hand to enhance depth perception during freehand interaction in a virtual reality (VR). Artificial fingertip deformation may generate mapping of real hand position and pseudo-haptics, improving UI usability. We conducted two experiments focusing on depth manipulation in both pointing and steering tasks. Our results revealed that changes in fingertip shape reduced operation time in pointing tasks and improved accuracy in steering tasks. Additionally, we conducted subjective evaluation surveys for both experiments, which showed improvements in pseudo-haptics, spatial perception, and user experience. Based on these results, we propose several applications and demonstrate that fingertip deformations in virtual hands can contribute to better 3D UI design.

CCS Concepts

• Human-centered computing → Virtual reality.

Keywords

Virtual reality, pseudo-haptics

ACM Reference Format:

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1 Introduction

Advances in Head-Mounted Display (HMD) have spurred the use of free-hand interaction using hand tracking with User Interfaces (UIs) in Virtual Reality (VR) [24, 55, 62]. This method is intuitive because it mimics real-world hand movements like grasping and touching, making manipulation in VR feel natural and simple. However, free-hand interaction faces difficulties in depth perception, and users tend to underestimate distances to objects [20, 32, 60]. This underestimation can cause premature contact recognition during UI interactions, degrading performance. Addressing this with haptic feedback [21] is challenging, as haptic devices may impede the freedom of hand movements and the problem of cost.

*Both authors contributed equally to this research.

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To address this issue, researchers have proposed visual feedback for UIs [35, 44] and pseudo-haptics techniques [45, 55]. However, previous methods may deteriorate performance due to the difference between the positions of real and virtual hands [14, 28, 49]. For instance, common penetration-blocking feedback causes such divergence: the virtual hand stops at a UI surface while the user's physical hand continues to move forward. Moreover, display techniques for showing path shadows during operations can support depth perception [15], but users must divide their attention between their own hands and the shadows. Despite prior studies to provide continuous depth cues through pseudo-haptic designs, they lead to these side effects; this is a research gap unresolved thus far.

We propose an interface using fingertip deformation as pseudo-haptics to improve depth perception in free-hand interaction with penetration-blocking UIs. When the virtual hand contacts an object, the fingertips deform depending on the penetration depth. Visual feedback through fingertip deformation and pseudo-haptics may allow users to perceive their spatial relationship with contacted UIs, while in continuous operations like dragging, synchronizing this deformation with pushing movements could potentially enhance depth perception. Such deformation can also induce illusory haptic sensations [43] by exploiting visuomotor discrepancies [4, 39].

In Experiment 1 (target-pointing), our method achieved performance comparable to conventional color-change feedback but scored significantly higher on subjective ratings for pseudo-haptics and spatial perception. In Experiment 2 (path-steering), the fingertip deformation improved manipulation accuracy even in high-difficulty conditions and was again rated significantly higher.

Our contributions are as follows: 1) We quantitatively showed that our method worked as a visual cue signifying contact for instantaneous interactions such as pointing, and as visual feedback with depth information for continuous interactions such as steering. 2) We qualitatively showed that fingertip deformation elicits pseudo-haptics sensations during UI interactions and improves spatial perception and user experience evaluations. 3) We proposed some applications based on the benefits of our method obtained from the experimental results.

2 Related Works

2.1 Freehand Interaction in VR

Freehand interaction encompasses indirect gesture-based approaches and direct manipulation techniques. Indirect gestures enable users to control virtual objects remotely through symbolic hand movements, without requiring direct contact between tracked virtual hands and target objects [33, 50, 51]. This contrasts with direct manipulation through virtual object contact [37, 42], where users interact by touching virtual elements with tracked hand representations, the focus of our work. Lacking real-world physical cues, freehand interaction can cause fatigue [10] and poor mid-air task performance. Moreover, underestimated depth perception in VR [25, 38, 60] often leads to premature contact and degraded UI performance. To address these challenges, many proposed solutions leverage visual feedback [12, 30, 34, 64]. For instance, Ownershift [24] reduced mid-air fatigue by offsetting the virtual hand from the real one. For delicate tasks, Zhu et al. [65] improved precision with small objects by zooming the interaction range via pinch gestures, while Park et

al. [44] investigated visual feedback on virtual buttons. To mitigate depth perception issues, Feld et al. [23] showed that hand occlusion in AR improves tapping operations. As a post-contact interaction, Tian et al. [56] used virtual fingertip brushes to access hidden UI windows through post-contact interaction.

2.2 Pseudo-Haptics in VR

Pseudo-haptics provides haptic sensations without dedicated hardware by creating a discrepancy between a user's actions and the corresponding visual feedback [4, 40]. This technique is well-suited for visually-driven VR environments, where it can create various perceptual illusions [5, 11, 46]. For example, manipulating the control-display (C/D) ratio between real and virtual hands affects the perceived weight and stiffness of objects [47, 59]. Pseudo-haptics is also effective for UI interactions; preventing virtual hand penetration when selecting buttons, for instance, induces kinesthetic feedback that improves performance and subjective ratings [8, 35, 36, 53].

However, pseudo-haptics for hand-based VR interaction faces challenges. Primarily, misalignment between real and virtual hands can degrade performance and user experience [14, 28, 49]. To address this, promising methods generate feedback via fingertip deformation [43] or attached visual guides [55] upon contact. Furthermore, Tian et al. [55] demonstrated that non-isomorphic approaches not only generate pseudo-tactile sensations but also improve interaction performance. Achibet et al. [3] proposed methods where passive haptics provides the underlying physical feedback while pseudo-haptics modulates the perceived characteristics. By providing this feedback without misaligning the hands, these techniques are suitable for performance-critical UI.

2.3 Summary

Pseudo-haptics approaches include rendering virtual brushes at fingertips [56] and deforming fingertips to simulate contact [43]. However, the quantitative impact of fingertip deformation on user task performance remains insufficiently validated. The novelty of our work lies in its comprehensive evaluation of a visual feedback method for enhancing depth perception through fingertip shape deformations, using both quantitative performance metrics and systematic subjective assessment for pointing and steering—fundamental HCI tasks.

We thus offer four hypotheses. H1: The proposed method improves performance in pointing operations. H2: The proposed method enhances performance in steering operations. H3: The proposed fingertip shape variation presents illusory haptic feedback to users, thereby improving subjective spatial perception. H4: Even when interacting with conventional VR UIs, the fingertip shape variation does not compromise users' sense of body ownership.

3 Deformation of virtual hand fingertip shape

Our system uses fingertip shape deformation as shown in Figure 1:(a) and (b). Experiments focused on index finger deformations only. The circular shape design of fingertip deformation was based on Morimoto et al.'s work [43]. Given human fingertips' inherent softness, integrating softness [54] into the deformation may elicit more natural kinesthetic feedback. In 2D UI interactions within VR environments, users frequently manipulate UI objects positioned

nearly parallel to themselves through gestures like those shown in Figure 1(b). To maximize the visual effect of fingertip deformation, our system implements circular deformations extending along the 'horizontal' and 'vertical' axes as viewed by the user.

Our system modifies a 3D model used for hand tracking within the Oculus Integration Unity Asset. Initially, the 3D model of the virtual hand was imported from Unity into the 3D modeling tool Blender. The thickness of the first joint of the index finger in the 3D model of the virtual hand was measured as 1.3 cm. Note that the measurements used in this experiment were independent of the dimensions of the hand of the user. The deformation of the fingertips was implemented using the Shape Keys feature of Blender, which allowed the shape to change from its original form to an arbitrary shape with animation. The deformation range is set using parameters from 0 to 100, allowing you to define everything from the unmodified state (parameter: 0) to any maximum deformation amount (parameter: 100). Additionally, adjusting these parameters enables to achieve shape deformations with change rates smaller than the maximum specified amount. Any maximum deformation amount was determined by measuring the change in the shape of the 3D modeling tool. The value in Blender was 1 = 1 meter.

This transformation was set to begin at the moment when the Unity system detects contact between the virtual object and fingertips of the virtual hand. The movement of the virtual hand in the VR space was synchronized with the movement of the real hand using hand tracking with a camera attached to the HMD. To prevent the virtual hand from passing through virtual objects, Poke Interaction from the Oculus Integration Unity Asset was used to detect contact between the virtual hand and virtual objects. The shape of the fingertip of the virtual hand deformed dynamically upon contact with the object. Particularly, when the user pushed the real hand in the vertical direction of the object, the fingertips of the virtual hand expanded in proportion to the pushing distance. In this process, hand speed or pressure does not affect the deformation. In our experimental system, the fingertip deformation change algorithm was designed to determine deformation based on the user's pressing distance relative to a predefined depth distance (D_I). The fingertip would not deform further even if the user moved their arm beyond D_I . The fingertip shape change parameter P is defined as $P = |F_p - T_p|/D_I$, where F_p and T_p are positions of the target being contacted and the fingertips, respectively. Finally, by inserting P into the variable of the Unity function `handMeshRenderer.SetBlendShapeWeight`, the rendering is completed within the app. For this study, the maximum scale factor for fingertip deformation in our system was set to 2.25, based on the findings of Morimoto et al. [43], who demonstrated that body ownership remains intact even when fingertips deform during interaction. Furthermore, artificial fingertip deformation may reduce UI visibility and decrease operability.

4 Experiment 1: Target Pointing

We investigated user performance in pointing tasks at varying depths to evaluate fingertip deformation effects. We compared three conditions: with deformation, without deformation, and with target color change. Color change served as a comparison since it provides common visual feedback. Previous research shows haptic

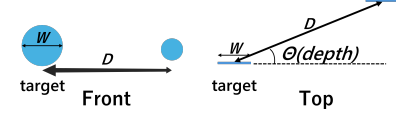


Figure 2: Setup for Experiment 1. Front: The front view of two blue targets. Top: The top-down view of the target placement. The width of the targets (W), distance between targets (D), and target depth angle θ ($Depth$) are the independent variables.

feedback improves VR depth perception [21], and pseudo-haptics using fingertip deformation may produce similar effects (H1).

4.1 Participants

12 university students (8 males, 4 females; ages 19-24, $M=21.9$) participated in this study. All had normal/corrected vision, no sensory impairments, and were right-handed. VR experience varied: 6 used VR intermittently over the preceding months, 4 monthly (1-2 times), and 2 daily (VR experts).

4.2 Task

Participants alternately pointed at targets of different depths (Figure 2). They were instructed to perform this task as quickly and as accurately as possible for ten selections. Upon completion, the targets disappeared and the task transitioned to the preparation phase for the next condition. The condition for determining that the target was correctly selected was limited to cases where the finger approached from the front side of the target, and the recording was made at the moment the finger touched and released the target. The user did not need to maintain the touch with the target for a fixed duration. An error was recorded when the participant touched the next target without touching the focused target. This error was detected when the fingertip entered the surrounding spherical area of a non-target, with a diameter twice that of the target. Selection when the fingertips were behind the target was recognized as invalid.

4.3 Design

We used a $3_{Visual} \times 7_{\theta} \times 2_D \times 2_W$ repeated-measures design. For *Visual*, we used three different visual feedbacks: *none* (no feedback), *finger* (fingertip expands based on penetration depth, maximum $2.25\times$ at 50 mm penetration), and *color* (target turns yellow upon contact). θ angles were positioned concentrically: -35° , -25° , -15° , 0° , 15° , 25° , and 35° relative to the left target, maintaining constant inter-target distance (see Figure 2). D was set to either 40 or 65 cm to ensure that both the left and right targets remained in the field of view throughout the task. W was set to either 2 or 4 cm in diameter of the target.

Each participant experienced 84 conditions ($3_{Visual} \times 2_W \times 2_D \times 7_{\theta}$), which were divided into three *Visual* sections, with task difficulty factors (W , D , and θ) randomized. *Visual* condition order was randomized across participants. Each condition required 10 selections, with the first selection for initiating a sequence. In total, we recorded $84_{conditions} \times 9_{selections} \times 12_{participants} = 9,072$ trials.

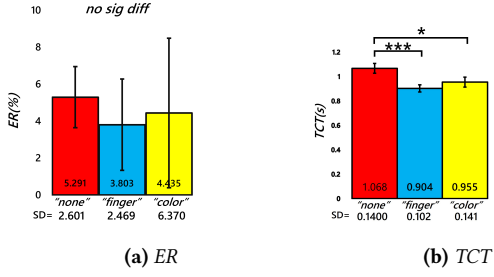


Figure 3: Results of ER and TCT for Experiment 1. Throughout this paper, error bars in the graphs indicate 95% CIs. Bar plots show the mean values, while standard deviations (SD) are indicated at the bottom of each graph. *** indicates $p < .001$, ** indicates $p < .01$, * indicates $p < .05$.

4.4 Apparatus

The VR system was developed using Unity (version 2021.3.41f). The setup comprised a laptop (Intel Core i7-12400, 32GB memory, NVIDIA GeForce RTX 3070 Ti) and Meta-Quest 3 HMD. Participants saw only virtual hands as their body representation, with no full avatar displayed. This equipment was also used in Experiment 2.

4.5 Procedure

Participants signed informed consent after receiving experiment explanations. The experiment used visual stimuli on an HMD in a quiet classroom to eliminate auditory and temperature influences. Participants sat in an unobstructed chair and received instructions on HMD operation, focus adjustment, and hand tracking. They then learned VR UI interaction tasks through practice sessions and completed the experimental tasks. Unlimited breaks were provided between tasks, with participants proceeding when ready. Total experiment duration was approximately 40 minutes per participant.

4.6 Measurement

The dependent variables were ER (error rate: percentage of trials with errors), TCT (task completion time for iterative selection), CT (contact time with target), and MT (movement time between targets). CT measures the time from the moment the fingertips touch the target until they are separated. The target was a flat circular area of diameter W (Figure 2). MT measures the time taken for the finger to leave the target and make contact with the next target. In addition to the speed and accuracy evaluation, we conducted subjective questionnaires for each *Visual* condition to measure pseudo-haptics illusion from fingertip deformation and evaluate UI interaction experience. Following previous research [35], we adopted elements of haptic illusion (Q1: "I felt the haptic sensation like touching the tangible object"; Q2: "It seemed as if I felt the touch of the button in the location where I saw the virtual button touched"), embodiment (Q3: "I felt as if the virtual hand was my own hand while pressing the button"; Q4: "It felt like I could control the virtual hand as if it was my own hand while pressing the button"), spatiotemporal understanding (Q5: "I could perceive the distance between my hand and the button well when pressing the button"; Q6: "I could perceive the exact timing when the button

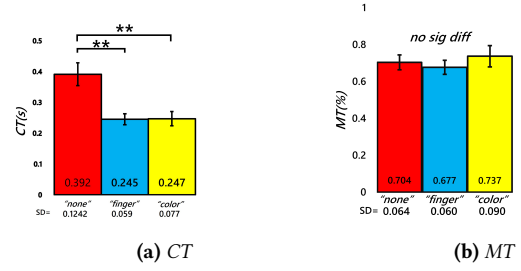


Figure 4: Result of contact time (CT) and movement time (MT) for Experiment 1. Average time per *Visual* is plotted. For CT, *finger* and *color* are significantly faster than *none*. MT, however, does not differ between *Visual* conditions.

would be pressed well"), and the short UEQ [49] (Q7: Pragmatic, Q8: Hedonic), evaluated on a 7-point Likert scale.

5 Result and Discussion of Experiment 1

Because ANOVA is robust against mild normality violations [41, 48, 57], we analyzed the data via RM-ANOVA ($\alpha = .05$). See supplementary materials for detailed ANOVA results. For the post-hoc test, we analyzed the data via pairwise tests with Bonferroni p-value adjustments.

Regarding outlier treatment, since TCT performs analysis on continuous operations, no outlier processing was implemented during analysis. For individual operations ER, CT, and MT, data where the absolute error of each trial from the mean operation time was 3SD [52] or more were excluded from the analysis as outliers. As a result, 3 trials (0.033%) were excluded from the analysis.

5.1 Quantitative Measures

5.1.1 Error Rate. In total, we observed 407 errors ($ER = 4.49\%$). The mean error rates were 5.291% for the *none*, 3.803% for the *finger*, and 4.453% for the *color* (Figure 3a). The mean ERs show lower ER for *finger* and *color*, but the significant main effect was not observed for *Visual*.

The absence of a significant main effect on error rate is likely attributable to a temporal mismatch. The instantaneous contact required by the pointing task provided insufficient time for users to perceive the continuous fingertip deformation feedback and incorporate it into their motor control.

5.1.2 Task Completion Time. For each condition consisting of ten selections, the number of pointing movements between targets was nine. Thus, an analysis of TCT was performed based on 9,072 trials.

Significant main effects were found on *Visual* ($F_{2,22} = 11.590$, $p < .001$, $\eta_p^2 = .513$), W ($F_{1,11} = 112.087$, $p < .001$, $\eta_p^2 = .911$), D ($F_{1,11} = 841.161$, $p = .005$, $\eta_p^2 = .987$), θ ($F_{6,66} = 2.390$, $p = .038$, $\eta_p^2 = .178$) and a significant $W \times D$ ($F_{1,11} = 10.180$, $p = .009$, $\eta_p^2 = .481$) interaction. From comparisons between *Visual* conditions, both *finger* ($p < .001$) and *color* ($p = 0.031$) showed significantly shorter TCT (mean and (SD) refer to Figure 3b), and there was no significant difference between *finger* and *color* (Figure 3b).

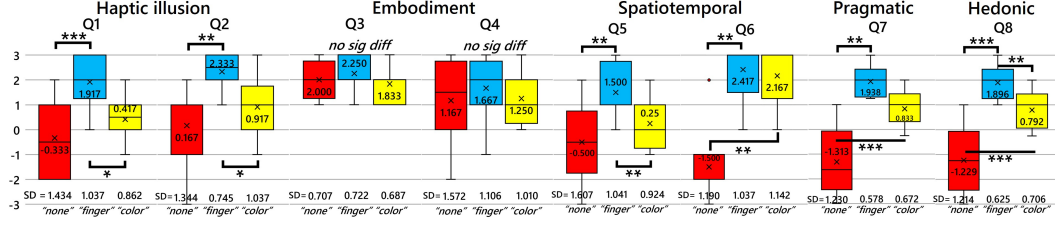


Figure 5: Results of subjective evaluation in Experiment 1.

The graph presents 7-point Likert scale ratings for haptic illusion (Q1, Q2), Embodiment (Q3, Q4), spatiotemporal perception (Q5, Q6), and pragmatic and hedonic qualities from the User Experience Questionnaire (UEQ). The *finger* showed significantly high scores for haptic illusions, spatial perception, and hedonic qualities.

These results show that *finger* potentially has a positive impact on pointing task performance, comparable to existing *color*.

5.1.3 Contact and Movement Times. For *CT*, an ANOVA on 9,670 non-error trials revealed significant main effects for *Visual* ($F_{2,22} = 15.559$, $p < .001$, $\eta_p^2 = .586$), *W* ($F_{1,11} = 6.323$, $p = .029$, $\eta_p^2 = .365$), and *D* ($F_{1,11} = 12.006$, $p = .005$, $\eta_p^2 = .533$), and a significant *Visual* $\times$ *W* ($F_{2,22} = 3.992$, $p = .033$, $\eta_p^2 = .266$) and *W* $\times$ *D* ($F_{1,11} = 5.637$, $p = .037$, $\eta_p^2 = .339$) interaction. Post-hoc comparisons on *Visual* showed that both *finger* ($p = .001$) and *color* ($p = .001$) conditions yielded significantly shorter *CT*s than the baseline (mean and (SD) refer to Figure 4a), with no significant difference between them (Figure 4a). These results were reasonable because the pointing task becomes more difficult as *D* increases and *W* decreases.

For *MT*, an analysis of 8,086 trials (excluding those adjacent to errors) revealed significant main effects for *W* ($F_{1,11} = 77.351$, $p < .001$, $\eta_p^2 = .875$) and *D* ($F_{1,11} = 444.230$, $p < .001$, $\eta_p^2 = .976$) and *Visual* $\times$ *W* ($F_{2,22} = 6.199$, $p = .007$, $\eta_p^2 = .360$) interactions. Post-hoc comparisons showed no significant difference between them (Figure 4b).

The reduction in *TCT* stemmed entirely from a lower *CT*. Both the *finger* and *color* conditions provided immediate contact feedback—unlike the ambiguous *none* condition—which allowed users to proceed to the next target faster. The *finger* feedback proved as effective for pointing performance as the conventional *color* method, validating it as a body-centric rather than a UI-centric feedback mechanism.

5.2 Evaluating Subjective and UEQ

We analyzed each questionnaire item ($n=36$ per question). Normality assumption was violated ($p < .05$), requiring non-parametric Friedman tests with Bonferroni-corrected post-hoc tests. Results appear in Figure 5. We found significant main effects for Q1, Q2, Q5, Q6, UEQ: Pragmatic, and UEQ: Hedonic¹. Post-hoc analysis revealed (mean and (SD) refer to Figure 5): for haptic illusion, significant differences between *none* – *finger* (Q1: $p < .001$, Q2: $p = .003$) and *finger* – *color* conditions (Q1: $p = .001$, Q2: $p = .012$). Embodiment questions showed no significant differences. Spatiotemporal perception showed significant differences for spatial perception

(Q5) between *none* – *finger* ($p = .007$) and *finger* – *color* ($p = .004$), and temporal perception (Q6) between *none* – *finger* ($p = .001$) and *none* – *color* ($p = .003$). UEQ pragmatic quality differed significantly between *none* – *finger* ($p = .002$) and *none* – *color* ($p = .006$). Hedonic quality showed significance across all pairs: *none* – *finger* ($p < .001$), *none* – *color* ($p < .001$), *finger* – *color* ($p = .004$).

Concerning the enhancement of spatial perception, it can be deduced that the direct visualization of the user's actions as a continuous quantity subsequent to contact with the target provided more intuitive and comprehensible feedback. This characteristic not only visualizes the sense of distance until contact but also the amount of hand movement after contact, potentially expanding the range of post-contact interactions.

6 Experiment 2: Path Steering

We investigated the user performance in steering tasks involving mid-air UI interactions. We assume that the depth perception by fingertip deformation shown in Experiment 1 is also useful for perceiving hand's depth during continuous tasks on a window (H2). However, continuous deformation may increase cognitive load and interfere with performance. We therefore tested how continuous fingertip deformation affects both planar task performance and depth control.

6.1 Participants

12 university students (9 males, 3 females; ages 20-23, $M=21.4$) participated in this study. All had normal/corrected vision, no sensory impairments, and were right-handed. VR experience varied: 7 used VR intermittently over the preceding months, 3 monthly (1-2 times), and 2 daily (VR experts).

6.2 Task

We used a sine-wave steering task, which is more challenging than linear steering and enables fair comparisons between different experimental conditions [31]. Participants moved fingertips left-to-right while maintaining contact with the displayed white path. Three error types were recorded: steering errors (deviation from path), lift errors (finger leaving window), and penetrating errors (excessive depth penetration). Lift errors occur when users release their fingers from the window in the -z direction (towards the user), while penetrating errors occur when users push their

¹Details are described in Appendix A.5

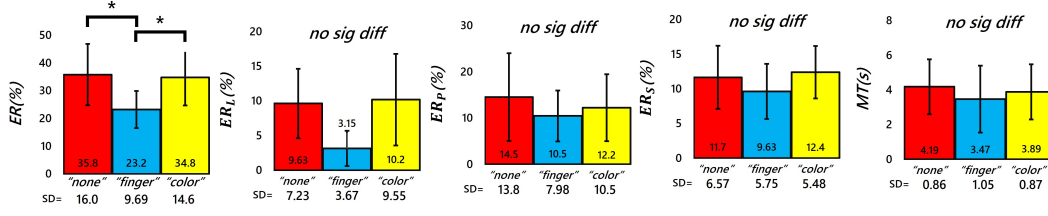


Figure 6: The results of ER , ER_L , ER_P , ER_S , MT between Visuals in Experiment 2. The *finger* condition showed significantly higher performance for ER . It also showed the higher performance for the other parameters, though this was not statistically significant.

arms beyond the predetermined limit in the z direction. This design required users to maintain fingertips within specific depth ranges while performing x - y directional tasks on a single window. We tested penetrating error distance (*Depth*) as an independent variable (detailed in Design section). Windows became transparent when penetration errors occurred, providing visual feedback.

6.3 Design

We used a $3_{Visual} \times 3_{Depth} \times 2_D \times 3_W$ repeated-measures design. *Visual* conditions were *none*, *finger*, and *color* (as Experiment 1), but with different meanings due to continuous window contact during steering. When *finger* was in contact with the window and had its fingertips within the path, the finger's shape changed continuously in response to the push-in Distance applied after the virtual hand's fingertips made contact with the window. *color* changed the path's color to yellow when the finger was in contact the window, providing binary contact information. Penetrating error thresholds (*Depth*) were 35, 45, and 70 mm; path lengths D were 365 and 730 mm; path widths W were 30, 45, and 60 mm. Fingertip deformation parameters matched Experiment 1 (maximum $2.25\times$ deformation).

We set D s so that the start and end positions were located at the peaks of the waves. The *wavelength* and *amplitude* of the sinewave were fixed at 220 mm and 70 mm for all conditions.

The order of the 3 visual conditions was balanced among 12 participants. A task set comprised all combinations of $2D$ and $3W$ presented in random order, with 5 such sets, all of which were performed for each *Depth* condition. The order of the *Depth* conditions was randomized. Completion of $2_D \times 3_W \times 5_{set} \times 3_{Depth} = 90$ trials in one *Visual* condition, then moved to the next *Visual* condition. So, each participant performed $90 \times 3_{Visual} = 270$ trials, and $270 \times 12_{participant} = 3240$ trials were recorded. If a trial failed, an error was recorded, and they moved to the next trial.

6.4 Procedure

The procedures for obtaining informed consent, explaining the experimental environment, and providing instructions for the HMD were the same as in Experiment 1.

After a few minutes of practice and confirmation of visual feedback, the task was started. Participants could start a trial by touching the start button displayed on the left side of the white path and then made a stroke in the path rightward. While stroking, the fingertip left a green trace. Participants were instructed to complete the task as fast and as accurately as possible. They were encouraged to take

a break at any time between trials, whenever they felt tired. Regardless of *Visual*, we provided identical visual error feedback when the fingertip deviated from the path. The duration of the experiment was approximately 45 min per participant.

6.5 Measurement

The position of the fingertip was recorded with a time stamp during the trials at a 72 Hz sampling rate. The dependent variables were ER , ER_S , ER_L , ER_P , MT , SD_W , and SD_D . ER is the overall error rate and represents the percentage of trials in which some error occurred. ER_S is the steering error rate, which is the percentage of trials in which deviation from the path occurred. ER_L is the lift error rate, which is the percentage of trials in which lift from the window occurred. ER_P is the penetrating error rate, which is the percentage of trials in which the fingertip penetrated too deeply beyond the window. MT is the time taken to stroke the path. SD_W is the standard deviation of the virtual hand fingertip coordinates in the task width direction. SD_D is the standard deviation of the real hand fingertip coordinates in the depth direction. Similar to Experiment 1, we also conducted psychophysical questionnaires in Experiment 2.

7 Result and Discussion of Experiment 2

Because ANOVA is robust against mild normality violations [17, 41], we analyzed the data via RM-ANOVA ($\alpha = .05$). See supplementary materials for detailed ANOVA results. For the post-hoc test, we used pairwise tests with Bonferroni p -value adjustments.

For the analysis, outlier detection was performed using the same criterion as in Experiment 1, where trials with absolute errors of 3SD or more from each participant's mean movement time in each condition were considered outliers. However, no outliers were detected using this criterion. A method for detecting outliers based on trajectory coordinates was also considered [52]. However, this coordinate-based outlier criterion was not adopted because trials end midway through the path in cases of Lift Error and Penetrating Error, meaning that outlier criteria might differ depending on the type of error. Therefore, the analysis was conducted on all recorded data. This is consistent with previous studies that conducted analyses without outliers in steering tasks [2].

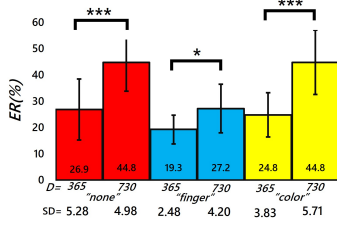


Figure 7: Visual and D interaction in ER. Increased errors due to D are reduced under finger condition.

7.1 Error Rate

7.1.1 Overall Error Rate. In total, we observed 1014 errors ($ER = 31.3\%$). We found significant main effects for all independent variables *Visual* ($F_{2,22} = 6.732, p = .005, \eta_p^2 = .380$), *Depth* ($F_{2,22} = 13.585, p < .001, \eta_p^2 = .553$), *D* ($F_{1,11} = 115.372, p < .001, \eta_p^2 = .913$), and *W* ($F_{2,22} = 55.972, p < .001, \eta_p^2 = .836$). Comparison between *Visuals* showed that *finger* had the lowest ER, with significant differences between *none* and *finger* ($p = .032$), and between *finger* and *color* ($p = .020$) (Figure 6a). There was no significant difference between *none* and *color*. These results showed that continuous visual feedback through fingertip deformation could improve the success rate of steering tasks performed on a single window. The main effects of the other parameters, that the ER decreased as *Depth* increased, as *W* increased, and as *D* decreased, are almost self-evident, since the ER decreases as the task becomes easier.

Significant interaction was observed only for *Visual* $\times$ *D* ($F_{2,22} = 5.125, p = .015, \eta_p^2 = .318$). The result shows that the increase in ER due to *D* is small only in *finger* (Figure 7). Increasing *D* required more lateral movement, which in some cases required stroking farther away from the body. In this case, the fingertip deformation might support the manipulation by visually indicating the possibility of depth direction errors.

7.1.2 Lift Error Rate. We observed 248 lift errors ($ER_L = 7.65\%$). Significant main effects were observed for *Visual* ($F_{2,22} = 3.577, p = .045, \eta_p^2 = .245$), *D* ($F_{1,11} = 13.588, p = .004, \eta_p^2 = .553$), and *W* ($F_{2,22} = 14.192, p < .001, \eta_p^2 = .563$). Comparison between *Visuals* showed that *finger* had the lowest ER_L , but no significant differences between *none* and *finger*, and between *finger* and *color* (Figure 6b). This result suggests that fingertip deformation feedback may improve depth perception by allowing users to use a slightly deformed finger as a visual guide. For the main effect of *W*, ER_L increased as *W* increased. It suggests that as on-plane tasks become easier, the users' focus on speed can lead to errors that cause the finger to lift off the window. For the main effect of *D*, ER_L increased as *D* increased, which was as expected.

Significant interactions were found for *Visual* $\times$ *W* ($F_{4,44} = 2.604, p = .049, \eta_p^2 = .191$), *D* $\times$ *W* ($F_{2,22} = 4.122, p = .030, \eta_p^2 = .273$), and *Depth* $\times$ *D* $\times$ *W* ($F_{4,44} = 3.746, p = .010, \eta_p^2 = .254$). For *Visual* $\times$ *W*, under *finger* condition, the increase in ER_L due to the increase in *W* is minimized (Figure 8). This indicates the possibility that fingertip deformation can improve the problem of users making errors by focusing on speed as *W* increases. That is, since the user

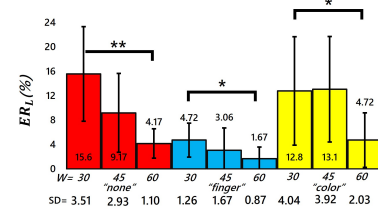


Figure 8: Visual and W interaction in ER_L . Increased errors due to *W* are reduced under finger condition.

can control the depth from the information of fingertip deformation, users may be encouraged to operate carefully to control depth, even in easy planner tasks. For *D* $\times$ *W*, the increase in ER_L due to the increase in *W* is increased by the increase in *D*. It seems reasonable to interact with the errors due to *W*, since as *D* increases, the time spent working on the task increases, and the probability of an error simply increases.

7.1.3 Penetrating Error Rate. We observed 402 penetrating errors ($ER_p = 12.4\%$). Significant main effects were found for *Depth* ($F_{2,22} = 24.789, p < .001, \eta_p^2 = .693$), and *D* ($F_{1,11} = 5.002, p = .047, \eta_p^2 = .313$). The main effect of *Depth* is obvious since this value is the threshold for penetrating error. For the main effect of *D*, ER_p increased as *D* increased, which was as expected. For *Visual*, there was no significant main effect, but the *finger* condition showed the lowest ER_p (Figure 6c). This result indicates that fingertip deformation feedback tends to reduce ER_p but has no statistically significant effect. This could be due to the difficulty in ascertaining the upper limit of fingertip deformation, which may not have had a significant effect on the ER_p .

Significant interactions were found for *Visual* $\times$ *W* ($F_{4,44} = 2.884, p = .033, \eta_p^2 = .208$), and *D* $\times$ *W* ($F_{2,22} = 3.980, p = .033, \eta_p^2 = .266$). The effect of *W* in each *Visual* condition was not consistent. Thus, additional experiments with more conditions would be needed to interpret these results. The inconsistent interaction effect for *Visual* $\times$ *W*, is difficult to interpret. One potential cause is that interruptions from lift and steering errors created an inconsistent number of opportunities for penetration errors to occur across conditions.

7.1.4 Steering Error Rate. We observed 364 steering errors ($ER_s = 11.2\%$). There were no significant main effects for *Visual* ($F_{2,22} = 1.410, p = .265, \eta_p^2 = .114$). While there were no significant differences, the *finger* condition had the lowest ER_s (Figure 6d). This could be due to the ease of depth control, which allows the user to focus on the on-plane steering task. At least, the continuous fingertip deformation does not seem to interfere with the accuracy of the on-plane steering task. Significant main effects were observed for *D* ($F_{1,11} = 25.579, p < .001, \eta_p^2 = .699$), and *W* ($F_{2,22} = 49.773, p < .001, \eta_p^2 = .819$) and a significant interaction was found for *D* $\times$ *W* ($F_{2,22} = 9.901, p = .001, \eta_p^2 = .474$). These results were reasonable because the steering task becomes more difficult as *D* increases and *W* decreases.

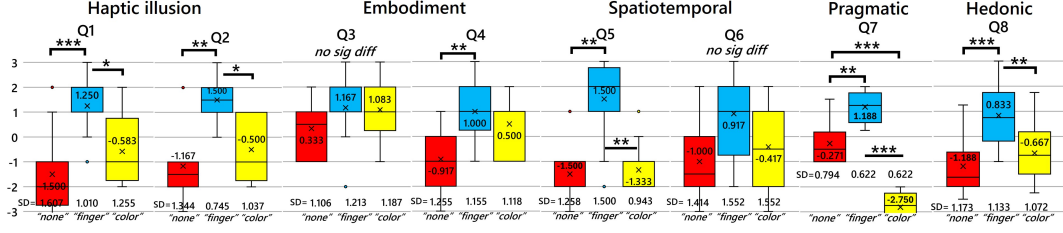


Figure 9: Results of subjective evaluation in Experiment 2. The graph presents 7-point Likert scale ratings for haptic illusion (Q1, Q2), Embodiment (Q3, Q4), spatiotemporal perception (Q5, Q6), and pragmatic and hedonic qualities from the User Experience Questionnaire (UEQ). The *finger* showed significantly high scores for haptic illusions, spatial perception, and UEQ questions.

7.2 Movement Time

To investigate the impact of fingertip deformation visual feedback on usability, we analyzed the mean movement time MT , without error trial data. As a result, there were 8 participants who made errors on all repeated trials under some conditions and had missing data. Therefore, we applied listwise deletion to ensure complete data for analysis. Then, data analysis was performed with 832 trials, which was the successful data of 4 participants².

Significant main effects were found on *Depth* ($F_{2,6} = 5.893$, $p = .038$, $\eta_p^2 = .663$), D ($F_{1,3} = 54.843$, $p = .005$, $\eta_p^2 = .948$), and W ($F_{2,6} = 51.124$, $p < .001$, $\eta_p^2 = .945$). There was no significant main effect for *Visual* (Figure 6e). According to the Finger Fitts Law [9], touch ambiguity at the fingertip decreases the movement time for finger manipulations compared to a stylus or cursor. Hence, the fingertip enlargement could worsen the MT . However, this effect does not seem to exist statistically. Rather, the overall trend is that the *Finger* condition shows the lowest MT . This indicated that fingertip deformation did not worsen the speed of operation.

A significant interaction was observed for $D \times W$ ($F_{2,6} = 35.327$, $p < .001$, $\eta_p^2 = .922$). This result is reasonable because the MT results fit the index of difficulty (*none*: $r^2 = .966$, *finger*: $r^2 = .964$, *color*: $r^2 = .929$), calculated by D/W based on the steering law [1].

7.3 Evaluating Subjective and UEQ

We analyzed 36 responses per questionnaire item using the same method as Experiment 1 (Figure 9). Friedman tests revealed significant main effects in Q1, Q2, Q4, Q5, UEQ: Pragmatic, and UEQ: Hedonic³. Post-hoc analysis revealed (mean and (SD) refer to Figure 9): for haptic illusion, significant differences between *none*–*finger* (Q1: $p < .006$, Q2: $p = .004$) and *finger*–*color* conditions (Q1: $p = .012$, Q2: $p = .007$). Embodiment showed no significant differences, including Q3 (body ownership). Agency (Q4) differed significantly *none*–*finger* conditions ($p = .005$). Spatiotemporal perception showed significant differences for spatial perception (Q5) between *none*–*finger* ($p = .003$) and *finger*–*color* ($p = .003$). Temporal perception (Q6) showed no significant differences. UEQ pragmatic quality differed significantly across all conditions (*none*–*finger*: $p = .003$; *none*–*color*: $p < .001$; *finger*–*color*: $p < .001$). Hedonic

quality differed significantly between *none*–*finger* ($p < .001$) and *none*–*color* conditions ($p = .003$).

The enhanced agency (Q4) from *finger* likely stems from the alignment between visual feedback and user intent in continuous tasks. This aligns with findings that congruent virtual hand control enhances agency and ownership [13], an effect our depth-mapped deformation successfully replicated.

8 General Discussion

8.1 Effect of Fingertip Deformation on Usability

Our method is shown to work as a visual cue signifying contact for instantaneous interactions such as pointing, and as visual feedback with depth information for continuous interactions such as steering. The pointing experiment results showed that the fingertip deformation reduced the contact time with the UI, similar to the visual cue using color. Therefore, the visual feedback of the initiation of fingertip deformation may have served as a visual cue for the contact. This provides partial support for H1 regarding performance in performance regarding operation speed. The steering experiment results showed that the fingertip deformation has a significantly decreased error rate than no visual feedback or visual feedback of color change. This indicates that the continuous fingertip deformation improved the user’s operating accuracy by providing information about the depth of the user’s real hand. This supports the accuracy aspect in H2. However, for movement time, the sample size was insufficient, resulting in low statistical power and no statistically significant findings. Further experiments with larger sample sizes are needed to validate the results concerning movement time. Finally, because the feedback modifies the user’s virtual body rather than the UI, it remains consistent across different interfaces, potentially reducing user confusion.

In subjective evaluations, both experiments demonstrated significant effects for the proposed method across haptic illusion, spatiotemporal perception, and UEQ items. This supports hypotheses H3 and H4. Both experiments showed the potential of using changes in virtual hand fingertip shape to present pseudo-haptics. This phenomenon may be attributed to the synchronized deformation of the virtual fingertips in conjunction with user movement, integrating visual and motor cues of “applying force to an object” in the user’s brain, thereby creating an impression of pressure on the fingertips. This finding is consistent with previous studies [43, 58], which showed that the deformation of the manipulated object can generate

²We recognize that the reliability of MT analysis is not high due to the high data missing rate, but the main contribution of our paper is the error reduction effect (see Section 8.3 for details).

³Details are described in Appendix B.3

haptic illusions. Haptic stimulation is commonly used to enhance UI performance in VR [7, 16, 19, 22, 63], and in this experiment, the pseudo-haptics feedback may have partially complemented physical haptic sensations, which could improve performance [22].

The proposed method achieved significantly higher UEQ evaluations compared to conventional methods. Hedonic quality scores were particularly elevated, indicating virtual hand fingertip deformation enhances users' pleasurable experiences. Following Hassenzahl's user experience model [29], the visual surprise and novelty of fingertip deformation contributed to this improvement. Since finger deformation when touching objects rarely occurs in reality, experiencing such visual effects in VR brought positive surprise and enjoyment to users.

Meanwhile, different outcomes were observed across experiments concerning pragmatic quality. In our steering task, significantly higher evaluations were shown compared to the color change condition; however, in our pointing task, this difference was not as pronounced. This correlates with performance improvement effects - while continuous feedback is more effective in steering, color changes alone were sufficient to induce noticeable performance improvements (increased accuracy) in pointing tasks.

8.2 Limitations and Future Work

In Experiment 2, data loss was observed in the analysis of *MT*. To address missing data, we piloted a re-experiment with four subjects where trials continued until success. This approach resulted in 89–247 trials per participant within 45 minutes. Thus, the experiment would likely take more than twice as long for most participants compared to Experiment 2, and participants reported significant fatigue accumulation. We considered reducing trial repetitions, but this would compromise the reliability of *MT* and *ER* [1, 61]. Consequently, we decided against re-running the experiment. While we performed an ANOVA on the four subjects who provided complete, valid data, we do not consider this our primary statistical analysis. On the other hand, the number of participants who provided valid data for each *Visual* condition was seven for *none*, ten for *finger*, and nine for *color*. This indicates that the *finger* was the most conducive to completing this demanding task, which supports its effectiveness in reducing errors.

The method's reliance on observing the virtual hand limits its utility in tasks where users' attention is elsewhere. Furthermore, pseudo-haptic perception varies individually [6, 18, 26].

The proposed system is technically implementable in AR/MR environments by superimposing the virtual hand onto the user's real hand. However, studies have shown that maintaining mixed perception with real objects reduces body ownership for the virtual hand [27], and in our proposed system, the additional deformation of the body may further diminish this sense of ownership.

9 Applications

The instantaneous contact detection from fingertip deformation has several applications in VR. For example, it allows users to immediately proceed to the next action after selecting a target (Figure 10a, b). This feedback can also provide a more immersive boundary warning than conventional visual grids by deforming the virtual hand as it exits the play area. Evoking the sensation of physical

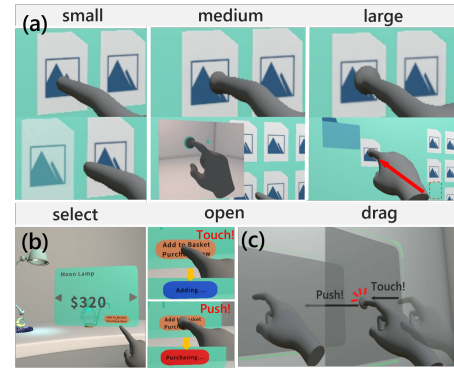


Figure 10: Applications for depth manipulation with finger deformation. This allows for (a) shortcut operations, (b) mapping of depth operations to multiple operations, and (c) access to specific windows floating in the air.

touch may also serve as a stronger safety cue that motivates users to return.

Fingertip deformation enables pressure-sensitive actions by distinguishing light touches from deep presses (Figure 10a). This provides a flexible alternative to time-based long-presses in confirmation dialogues (e.g., in-game purchases) that prevent accidental inputs. Users can either press deeply for immediate confirmation or use a traditional long-press, enhancing operational flexibility based on preference and context.

Our method also enhances 3D folder operations (Figure 10b). Traditional 2D metaphors (e.g., tap, double-tap, press-and-hold) are ill-suited for mid-air VR interaction, where precise depth perception is challenging. By mapping distinct actions like select, open, and drag to the user's press depth, our method provides a seamless and intuitive alternative.

Furthermore, our approach can be applied to multi-layered window UIs (Figure 10c). Users are familiar with overlapping windows from the conventional desktop metaphor. By applying this metaphor to VR mid-air interaction, users can intuitively select and manipulate windows across different layers based on press depth. For example, users can light press to interact with the foreground window, while deeper press to interact with the background one.

10 Conclusion

We investigated the effects of virtual fingertip deformation on user interaction in VR through pointing and steering experiments. Results show that fingertip deformation enhances depth perception and provides pseudo-haptics feedback for freehand interaction. While pointing task times were comparable to conventional visual feedback, our method significantly improved accuracy in steering tasks with depth constraints. This suggests fingertip deformation can serve as a visual contact cue, and its continuous application aids depth perception when manipulating 2D interfaces. Our method received higher ratings for (e.g.) pseudo-haptics and spatial perception compared to traditional methods. Our approach uses visual deformation to provide intuitive depth cues and pseudo-haptics feedback, contributing to more effective VR interaction.

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